animal sacrifices may have been performed; at Harappa, the platform was twenty-three feet high. At the site of Kalibangan, in northwestern India, archaeologists uncovered a pit on top of the platform containing burned cattle bones.



FIGURE 3.25 The Citadel Mound at Mohenjo-Daro. The stupa at Mohenjo-Daro, located in present-day Pakistan, was built circa 2500 BCE. The identification and dating of the stupa in 1923 by Indian archaeologist Rakhal Das Banerji led to the first major excavations at the site in the 1920s and 1930s. (credit: "Stupa at Mohenjo-Daro" by Omair Anwer/Flickr, CC BY 2.0)]

The citadels also included public baths. At Mohenjo-Daro, the tank of the bath was forty feet long and twenty-three feet wide and entered by staircases on either side. A nearby well provided the water. Archaeologists have uncovered a large hall supported by pilasters composed of mud-brick at Mohenjo-Daro, as well as a multistory residence built around an open courtyard. This evidence suggests that ritual specialists, perhaps priests, lived and performed religious functions in the citadels that may have required them to bathe in the large public bath and congregate in the hall. An extensive granary for grain storage was found at Mohenjo-Daro. Farmers from outlying rural areas undoubtedly produced the surplus crops that were stored here to provide sustenance for the elite, religious specialists, and other city residents such as merchants and artisans, just as in Mesopotamia and Egypt.

The lower sections of the Indus valley cities consisted of residential quarters and workshops. At Mohenjo-Daro and Kalibangan, the houses typically consisted of four to six rooms built around a central courtyard and were equipped with wells to provide running water to a bathroom. Larger homes in the cities were multistory with as many as thirty rooms. There is also evidence of devices attached to some of the roofs that pumped wind into homes and other large buildings to cool them.

LINK TO LEARNING

Some of the most intriguing evidence at the large Indus valley civilization sites indicates the residents had sophisticated knowledge of water engineering and built citywide drainage systems (https://openstax.org/l/77IndusDrainage) with covers for servicing. Evidence was found of indoor toilets that connected to this drainage network (https://openstax.org/l/77DrainageNet) that ran throughout the city.

The Indus valley cities also included an industrial area where workshops were located. At Harappa, this district included quarters for the laborers who worked there. The ancient city at the site of Lothal near the Indian Ocean included a dockyard and a warehouse for incoming trade goods. The inhabitants of these cities may have included the artisans and merchants who provided goods for the ruling elite. As in ancient Mesopotamia and Egypt, the majority of the population were probably farmers who lived in the outlying rural areas surrounding each city.

Trade, Writing, and Religion

Archaeological work has revealed that a considerable amount of trade flowed between Mesopotamia and the Indus valley. Cuneiform tablets from Mesopotamia refer to the Indus valley as *Meluhha* and document that precious stones such as lapis lazuli and carnelian, as well as marine shells from the Indian Ocean, were imported from there. Merchants traveled by sea across the Indian Ocean and by land over the Iranian plateau (Figure 3.26).

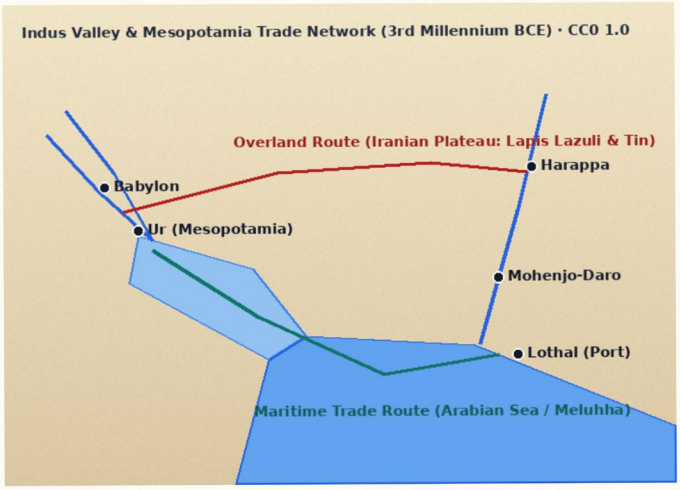


FIGURE 3.26 Indus Valley Trade with Mesopotamia. From the work of archaeologists, we know that significant trade took place between the inhabitants of the Indus River valley and their contemporaries in Mesopotamia. (credit: "Indus-Mesopotamia Trade Routes Map", CC0 1.0 Public Domain; text: Rice University, OpenStax, under CC BY 4.0 license)

The influence of Mesopotamia on the Indus valley culture is evident in the use of seals. The Indus valley seals were inscribed with depictions of human figures and animals such as bulls and goats, likely totems for families or lineages, and brief inscriptions that likely indicate names, titles, or occupations. Merchants marked ownership of goods by making an impression of the seal on the soft clay that covered the mouth of the jar or other vessel that held the objects, or on clay tags attached to sacks of grain (Figure 3.27). A similar practice occurred in contemporary Mesopotamia, where the seal was in the shape of a cylinder that could be rolled to leave an impression. Archaeologists have found seals from the Indus valley in ancient Sumerian cities such as Ur.



FIGURE 3.27 An Indus Valley Seal. On the left is a seal from the Indus valley civilization, and on the right is a modern clay impression made from it. Such seals typically showed animals and a script that remains to be deciphered. (credit: modification of work "Stamp seal and modern impression: unicorn and incense burner (?)" by Dodge Fund, 1949/The Metropolitan Museum of Art, Public Domain)

The script that appears on many of the seals is unique to the people of the Indus valley civilization, and scholars have yet to decipher it. It seems to consist of phonograms, signs for the sounds of syllables, and there appear to be about four hundred such signs. Many speculate that the language written in this script may be

related to the Dravidian languages still spoken across southern India today. It may also be similar to the script invented by the Elamites of southwestern Iran. Unlike the case in Mesopotamia and Egypt, archaeologists have not uncovered clay tablets or papyrus rolls in the Indus valley.

THE PAST MEETS THE PRESENT

Deciphering Indus Valley Script

One of the great mysteries surrounding Indus valley script is what exactly it represents. Was it a means of capturing spoken language, or did the marks simply indicate whether taxes had been paid on an item or signify the quality of a particular good (in the same way we use stars to rate products and services online today)?

Rajesh Rao, a professor of computer science at the University of Washington who became fascinated with the Indus valley civilization as a child in India, created a computer program to help him answer this question by measuring conditional entropy in Indus valley writing. Conditional entropy measures the degree of randomness in a sequence. In a system of writing that encodes language, there is a fairly low level of randomness. Letters appear frequently in some combinations and rarely or never in others. For example, in the English language, the letter *q* is usually followed by the letter *u* and never by the letter *k*. At the beginning of a word, the letter *h* is never followed by a consonant.

Rao tested the conditional entropy of the Indus valley script along with that of several natural languages including English, artificial languages such as those written for computers, and other sequences not related to language such as human DNA sequences. He discovered that the Indus valley script has a degree of randomness similar to that of natural languages, leading him to the conclusion that the symbols do represent a language and are not simply marks of quality or signs that something has been taxed. He also concluded that the rate of conditional entropy in Indus valley writing resembled that of Old Tamil, an earlier form of the Tamil language currently spoken in southern India that belongs to the Dravidian language family. This idea angered many Indians, especially those who speak Hindi, a language derived from Sanskrit that is the first language of many people in northern India. This issue is both historical and modern. If the language of the ancient Indus valley was Dravidian in origin, it calls into question the ancestry of the people who lived there, and given the evolution of the caste system, it raises questions about social identities that have existed for centuries. The issue remains in question, and the controversy about ancestral origins is far more complex than the single issue of language.

- Why do you think the origin of language matters so much to the people of India?
- If Indus valley writing were deciphered, what could historians learn about the culture that we cannot currently know?
- In what other ways could computers help historians learn about the past?

Bronze technology probably also entered the Indus River valley by the third millennium BCE through trade with Mesopotamia. Merchants from the Indus valley may even have exported tin from Afghanistan to Mesopotamia, since this metal was in demand for the manufacture of bronze.

Notwithstanding the many obvious Mesopotamian influences and trading connections, the people of the Indus valley civilization developed their own unique culture. Their distinctive religion may have shaped later cultures in India. For example, clay figurines from the Indus valley that are believed to depict deities are often interpreted as portraying a goddess whose female attributes are similar to those of the Hindu goddess Durga, consort of the god Shiva. Some seals depict a horned three-faced figure surrounded by animals, which closely resembles the Hindu deity Shiva when represented as the Lord of the Animals (Figure 3.28). Archaeologists have also found stones molded into shapes that resemble *lingams* and *yonis*, which are representations of male and female sexual organs associated with the worship of Shiva. The people of the Indus valley buried their dead with modest grave goods such as clay pots, which suggests a belief in an afterlife. However, there is

no evidence of temples or monumental tombs such as the Egyptian pyramids or the royal tombs of Ur in Mesopotamia.



(a) (b)

FIGURE 3.28 Possible Influence of Indus Valley Religious Imagery. Historians speculate that the religion of the Indus valley might have influenced later Indian cultures. For example, a seal from 2600–1900 BCE discovered at Mohenjo-Daro (a) shows a seated figure with three faces surrounded by animals. A much later sixth-century CE portrayal of Shiva from the Elephanta Caves in western India (b) shows the Hindu god seated, also with three faces and surrounded by animals. (credit a: modification of work "Shiva Pashupati" by Unknown/Wikimedia Commons, Public Domain; credit b: modification of work "Elephanta Caves" by Christian Haugen/Flickr, CC BY 2.0)

The artisans of the Indus valley created unique sculptures in clay, stone, and bronze. One of the more spectacular stone works from Mohenjo-Daro depicts a serene bearded man who may be a priest or leader in the community. A tiny bronze figure appears to represent a young woman dancing (Figure 3.29). Her bracelets and necklaces indicate that the people of this culture employed artisans to manufacture such adornment, which may have indicated high social status. Artisans of the Indus valley were also very busy manufacturing pottery and seals. Their artistic designs influenced artisans in neighboring cultures over a wide area, from the upper Ganges River valley in what is today northwestern India to Baluchistan in western Pakistan and southeastern Iran.